The Industrial Revolution in the Idea Factory

AI and the Research Workflow

Matt Webb

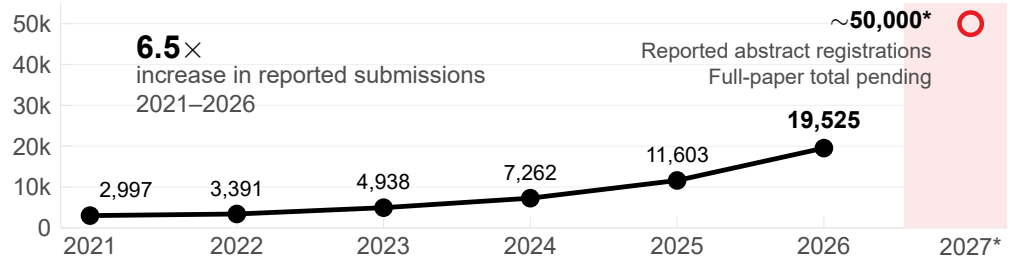
Carleton University

mattdwebb.github.io

September 2026

The submission pipeline is expanding rapidly

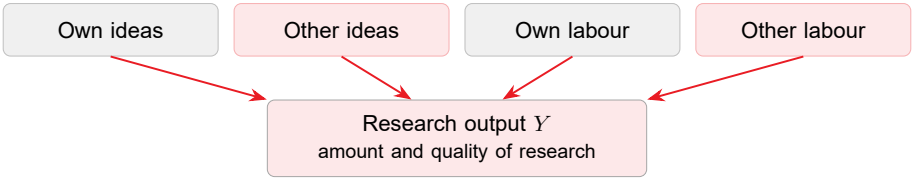
International Conference on Learning Representations (ICLR), 2021–2027



2027 is provisional: roughly 50,000 reported abstract registrations; the full-paper deadline is 25 September 2026. Submission IDs are not a verified count of valid papers.

Historical totals: ICLR 2021, 2022, 2024 fact sheet, 2026 fact sheet. 2027: author report; official deadlines. As of 21 Sep. 2026.

What goes into producing research?



$$Y = F(I_{\text{own}}, I_{\text{other}}, L_{\text{own}}, L_{\text{other}})$$

Each input takes time and human or robot capital (skills and tools). “Other” inputs can come from people or machines.

AI changes what these inputs can do and what they cost.

Conceptual framework / practical workflow proposed for this talk.

Research is already a team sport

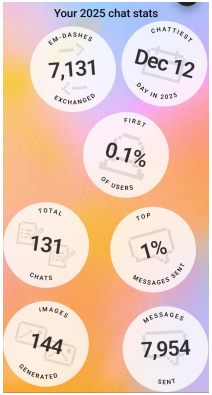
A synthetic acknowledgments footnote from a top-five economics paper

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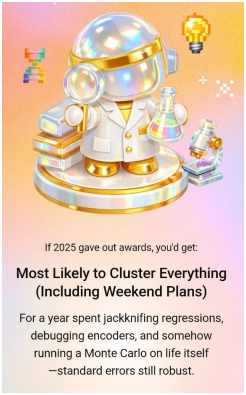
*We thank Daron Acemoglu, Joshua Angrist, David Card, Esther Duflo, Claudia Goldin, and Guido Imbens for generous discussions and insightful comments. We are grateful to the editor and three anonymous referees for suggestions that substantially improved the paper. We also thank seminar participants at Berkeley, Chicago, Harvard, MIT, Princeton, and Stanford, and participants at the NBER Summer Institute. Our research assistants provided excellent help with data construction and replication. All remaining errors are our own.

Synthetic example, not an actual acknowledgment. The named economists did not contribute to this talk.

I may not be a representative user…



First 0.1% of users

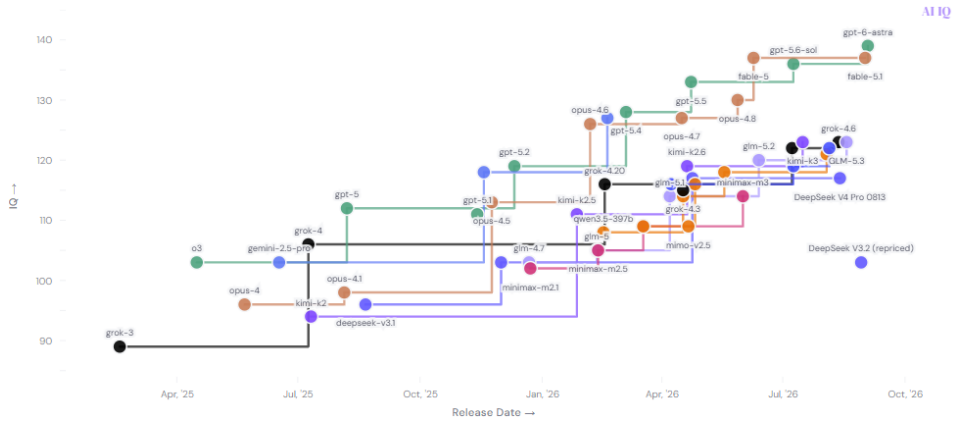


"Cluster everything"

Matt Webb's original screenshots. First 0.1% refers to joining, not usage intensity.

A recent snapshot of leading AI models

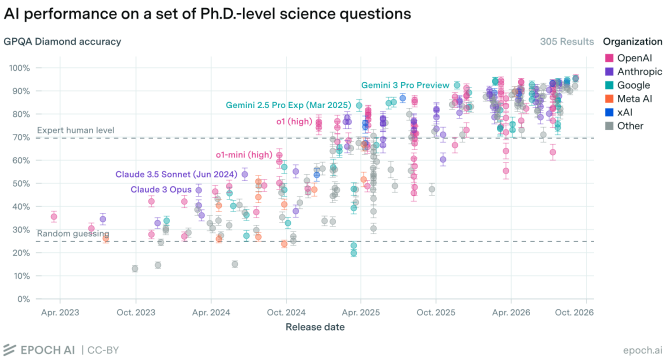
AI IQ / after Astra's release



AI IQ; user-supplied snapshot. A synthetic benchmark index, not a psychometric IQ measurement.

An exam rather than an IQ scale

GPQA Diamond / 198 graduate-level questions in biology, chemistry and physics



Knowledge on a test is not the whole research workflow.

Epoch AI, GPQA Diamond. Original graph export, 21 Sep. 2026. CC BY 4.0.

AI capability $\neq$ human + AI capability

Physicians / Goh et al. (2024)

Median (middle) diagnostic-reasoning score

Conventional resources	**74%**
Access to GPT-4	**76%**
GPT-4 alone	**92%**

AI access: adjusted gain 2 pp, 95% CI $[-4, 8]$. 50 physicians.

Public / Bean et al. (2026)

Relevant conditions identified in scenarios

Models tested alone	**94.9%**
People using those models	$< 34.5\%$

1,298 participants. Assisted users did no better than controls.

pp = percentage points; CI = confidence interval (a range expressing statistical uncertainty).

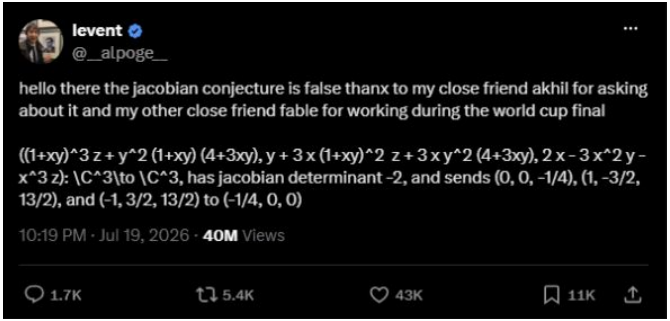
Capturing the capability requires a better human–AI workflow.

Different tasks and scores across studies. Model-alone results are separate benchmarks, not randomized human arms.

Goh et al. (2024), JAMA Network Open; Bean et al. (2026), Nature Medicine. Scenario studies, not real-world clinical outcomes.

From an announcement to something we can check

Jacobian conjecture (a proposed mathematical claim) / July 2026



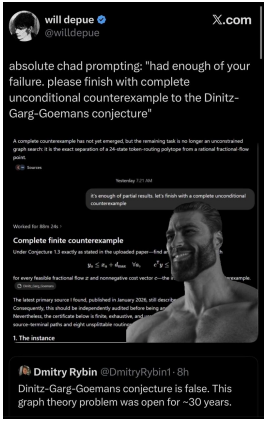
The reported example
A polynomial map $F : \mathbb{C}^3 \to \mathbb{C}^3$ (a rule using powers and sums), with constant, nonzero Jacobian determinant (local scaling), yet different inputs sharing an output.

The check
Compute the determinant and check the input points with exact arithmetic (no rounding).

The useful result is an example others can check independently.

Levent Alpöge, original X post (19 July 2026), captured 21 Sep. 2026. Exact symbolic check included in code/.

“Do a breakthrough"



“It’s enough of the partial results.”

Prompt wording normalized; original at left.

A short intervention preceded a counterexample (an example that disproves the claim) to the **Goemans cost conjecture**.

The finite example was later checked in Lean (a language where the computer checks each logical step).

Depue / Rybin prompt screenshot (July 2026). Independent Lean verification: cost-preserving rounding conjecture, not the Dinitz–Garg–Goemans theorem.

Navier–Stokes: discovery with a computer-checked proof

OpenAI / released 8 September 2026

Smooth motion can break down in finite time

Three-dimensional incompressible flow (constant density), initially at rest, with **smooth external forcing** (a smoothly varying push) and finite energy.

The result
A blow-up construction (a proof of speeds growing without bound), with a public proof written in Lean.

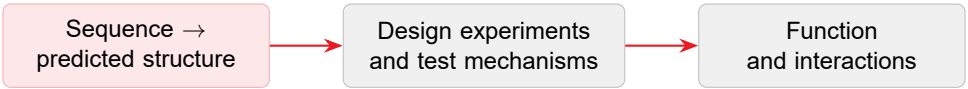
The research workflow
AI agents worked together on the argument, then wrote a version the computer could check.

A released theorem/proof claim, not just a simulation. This is the forced case; it does not prove blow-up without forcing.

OpenAI, announcement and paper (8 Sep. 2026); public Lean certificates. Status checked 24 Sep. 2026; certificate not rerun for this talk.

When structures get cheaper, the work moves

AlphaFold / the structural biologist's workflow



What became cheaper
Initial models, experiment planning, and interpreting crystallography / cryo-EM (X-ray / electron-microscope measurements).

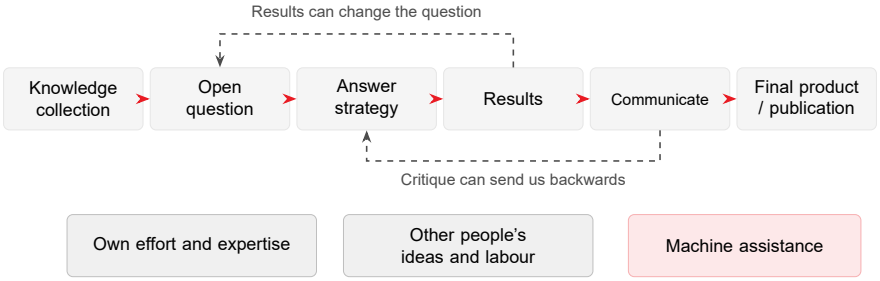
How researchers adapted
Investigate harder structures and what they mean for living systems. Test what proteins actually do.

Evidence: Yu (2026) finds research directions shifted and citations rose, without a significant rise in publication counts.

An abundant input changes which research tasks are valuable.

AlphaFold and the future of structural biology (2023); Yu (2026), CESifo WP 12462. Workflow synthesis; predictions are not experimental determinations.

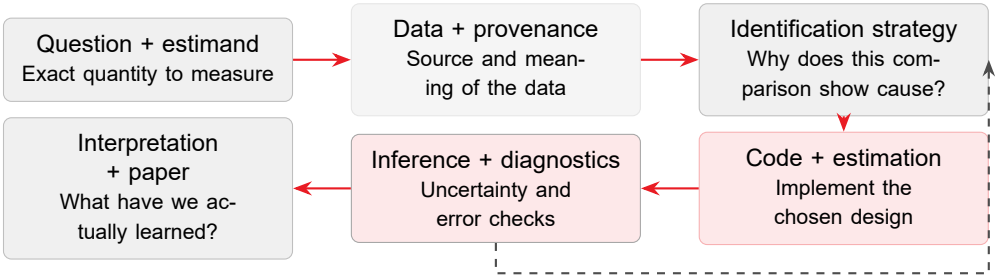
A map of the idea factory



Organize the tools around the work, not the other way around.

Conceptual framework / practical workflow proposed for this talk.

Research with data: the workflow

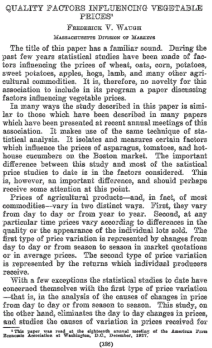


Making the code run is one box. It is not the research design.

Conceptual framework / practical workflow proposed for this talk.

A regression used to be a major undertaking

Waugh (1928) / hedonic regression: linking asparagus prices to quality



QUALITY FACTORS INFLUENCING VEGETABLE PRICES*

FREDERICK V. WAUGH

Massachusetts Division of Markets

The title of this paper has a familiar sound. During the past few years statistical studies have been made of factors influencing the prices of wheat, oats, corn, potatoes, sweet potatoes, apples, hogs, lamb, and many other agricultural commodities. It is, therefore, no novelty for this association to include in its program a paper discussing factors influencing vegetable prices.

In many ways the study described in this paper is similar to those which have been described in many papers which have been presented at recent annual meetings of this association. It makes use of the same technique of statistical analysis. It isolates and measures certain factors which influence the prices of asparagus, tomatoes, and hothouse cucumbers on the Boston market. The important difference between this study and most of the statistical price studies to date is in the factors considered. This is, however, an important difference, and should perhaps receive some attention at this point.

Prices of agricultural products—and, in fact, of most commodities—vary in two distinct ways. First, they vary from day to day or from year to year. Second, at any particular time prices vary according to differences in the quality or the appearance of the individual lots sold. The first type of price variation is represented by changes from day to day or from season to season in market quotations or in average prices. The second type of price variation is represented by the returns which individual producers receive.

With a few exceptions the statistical studies to date have concerned themselves with the first type of price variation—that is, in the analysis of the causes of changes in price from day to day or from season to season. This study, on the other hand, eliminates the day to day changes in prices, and studies the causes of variation in prices received for

*This paper was read at the eighteenth annual meeting of the American Farm Economic Association at Washington, D.C., December, 1927.

(185)

The question

How does observed quality affect market prices?

$$P_i = \alpha + \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$$

Schematic equation; not Waugh's original coefficients.

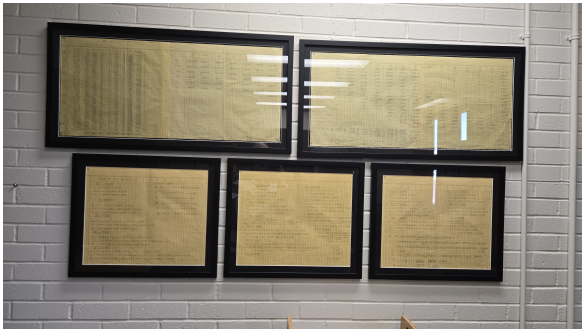
Measurement, data collection and arithmetic were all part of the work.

The scarce input was not always the same as the scarce input today.

Waugh (1928), Quality Factors Influencing Vegetable Prices, Journal of Farm Economics 10(2):185–196. Publisher's original first-page preview (p. 185).

What is still scarce after the arithmetic is cheap?

Then: computation worth framing



Framed hand-computed regressions in our department.

Now: a command

```
regress y x1 x2 x3
```

The calculation alone is no longer much of a research contribution.

The question, design, data and interpretation still matter.

My worry: is “producing a paper” starting to follow the same path?

Photograph: Matt Webb, supplied 22 September 2026. Original image preserved; shown without alteration.

Knowledge collection: ask for a map and its evidence

01 / knowledge collection

Use the model for
Definitions, related research, competing explanations, search terms, and a first map of the field.

Ask it to separate
What the source says; what the model concludes; what it does not know.

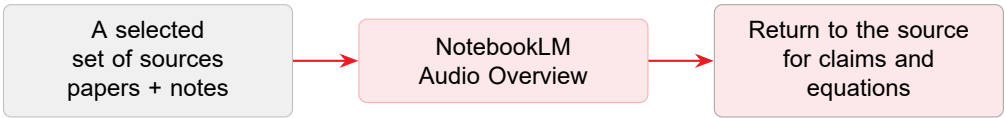
Try this prompt
Using only these papers, compare the exact quantities they aim to measure, their assumptions and their findings. Attach a page or section to each claim. Mark missing information rather than filling the gap.

A useful literature map is a route back to the sources, not a substitute for them.

Conceptual framework / practical workflow proposed for this talk.

Give your literature group an audio overview

01 / knowledge collection / NotebookLM audio summaries



Useful for orientation
What is the question? Where do the papers disagree? What should I read closely?

Not the last word
An engaging explanation can still omit a qualification or misstate a result.

Demo: generate an overview, then check one claim against the uploaded paper.

Google Help, Audio Overviews (NotebookLM / Gemini Notebook); interface and product names can change.

Use the model to make the question harder

02–03 / open question and answer strategy

Finding questions
What is missing? What is actually new? What existing result would make this unnecessary?

Choosing an approach
What assumptions does each approach require? What result would distinguish the explanations?

Try this prompt
Give me three plausible explanations for this pattern. For each, propose one empirical test and one reason that test might fail. Do not select a preferred story yet.

Ask for alternatives and failure conditions, not just encouragement.

Conceptual framework / practical workflow proposed for this talk.

From asking for an answer to delegating a task

Chat-based LLM

An LLM is a large language model: AI that works with text. You supply context, request an output, and carry it to the next step.

The user runs the loop
Ask → copy → run → ask again.

Agentic system (AI that takes action)

With tools and permissions, it can inspect files, search the project, edit, run code, test and revise.

The agent runs multiple steps
You specify the target and review the evidence.

Coding: Codex, Claude Code, Muse Code (Meta).

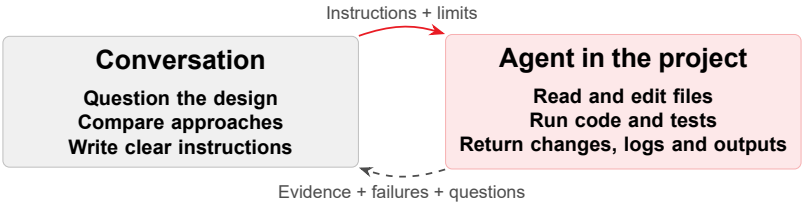
Broader task agents: Grok Bot and Muse, Meta's new personal agent.

The practical difference is tools, access and the ability to repeat work and checks.

Official documentation: Codex, Claude Code, Grok Bot, Muse (8 Sep. 2026), Muse Code. Checked 24 Sep. 2026.

The conversation and the agent work in tandem

03–04 / strategy into execution



The researcher owns the handoff and the decision to accept the result.

Examples: ChatGPT / other LLMs for discussion; Codex / Claude Code for work on project files.

Claude Code documentation; OpenAI coding-agent documentation. Feature availability depends on interface and plan.

Version control is your research undo history

Git saves named snapshots
A *commit* records a version. A *diff* shows exactly what changed. A *branch* keeps an experiment separate from stable work.

GitHub shares the project
Store the history online, collaborate and review proposed changes with other people.

Commit → delegate → inspect diff → test → keep/revert

Agents can make confident mistakes across many files. Commit before a large task, use a branch, and undo (revert) bad edits to files Git tracks.

More automated editing makes version control more important.

GitHub Docs, About GitHub and Git. Workflow advice proposed for research projects.

Give an agent a research brief, not a wish

04 / results / work someone else can rerun

Specify the target
Inputs, observations to include, target quantity, research design and expected outputs.

Specify the boundaries
Do not alter raw data. Do not change the model to obtain a preferred answer.

Specify the evidence
Tests, logs, a change summary and a clean rerun.

Try this prompt
Implement this specification in a new branch. Preserve raw data. Add automatic checks of which observations are included and how data files are joined. Run the tests. Report every deviation from the brief, and stop rather than silently changing the quantity being measured.

The deliverable is a checkable project, not a reassuring chat message.

Claude Code documentation; OpenAI coding-agent documentation. Feature availability depends on interface and plan.

Try to make the checks fail differently

Different methods can expose different mistakes

Change the checking method
Another model, separately written code, another programming language, or a hand-calculated special case.

Keep the research target fixed
Same observations, target quantity, weights and method for measuring uncertainty.

Why this helps
Agreement is more informative when the checks do not share the same failure mode.

Repeating the question is weaker
The same model can repeat the same convincing mistake.

Cunningham's framing: **"uncorrelated errors"**

Errors that do not tend to occur together. A goal for designing checks, not a guarantee of independence.

Cunningham (12 Mar. 2026), six-package audit. Different software cannot rescue a shared identification error.

Code to mathematics — and back again

04–05 / DiD (difference-in-differences): comparing changes over time between groups

The model written as an equation

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}$$

Unit and time effects account for fixed differences between units and common changes over time. The treatment indicator marks who is treated and when.

Ask the model to identify
Reference groups, who is included, weights and the method for measuring uncertainty.

The model written as code

```
regress y D i.id i.t,
  vce(cluster id)
```

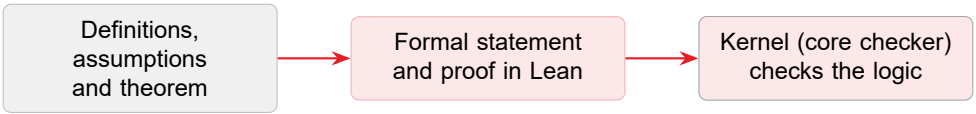
Illustrative two-way fixed-effects specification, not a recommendation for heterogeneous staggered-adoption DiD.

Translating back and forth can reveal differences between the paper and the code.

Conceptual framework / practical workflow proposed for this talk.

Lean: a second language for mathematical claims

A formal proof assistant (software that checks mathematical proofs)



AI can generate
Proposed definitions, proof steps and revisions after Lean reports errors.

Lean verifies
That the resulting proof establishes the exact statement from its axioms (starting assumptions).

Generation by AI. Verification by the proof assistant.

Humans still check that the formal statement means what was intended.

Theorem Proving in Lean 4: Axioms and Computation. Completion requires auditing placeholders and additional axioms.

Small jobs with a very high annoyance-to-value ratio

05 / LaTeX: software for typesetting equations and papers

Handwriting → LaTeX
Transcribe first. Mark ambiguous symbols. Check the image against the result.

Symbol check
Find every use of G, g, T and t. List what each symbol means. Flag conflicting definitions.

Code ↔ equations
Describe what runs, not what the comments say should run.

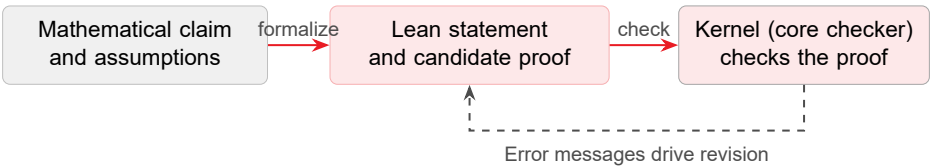
Try this prompt
Check the symbols only. Do not rewrite the argument. List each inconsistency with the location, current usage and a proposed minimal change.

Narrow instructions often make it easier to check the answer.

Conceptual framework / practical workflow proposed for this talk.

Generation and verification can be different jobs

04 / Lean and formal proofs



Audit proof placeholders (`sorry`) and added axioms before calling a proof complete.

A convincing explanation is not a checked proof. `sorry` marks an unproved step.

Theorem Proving in Lean 4: Axioms and Computation. Formal verification is conditional on the encoded statement and axioms.

Fermat's Last Theorem gets a computer-checkable form

A known theorem, now written as a proof in Lean

$$a^n + b^n \neq c^n \qquad (a, b, c > 0,\ n > 2;\ a, b, c, n \in \mathbb{N})$$

1995

Wiles and Taylor–Wiles, building on Frey, Serre and Ribet.

2026

Anthropic released a complete AI-assisted Lean proof on 4 September, built on Mathlib (a shared mathematics library).

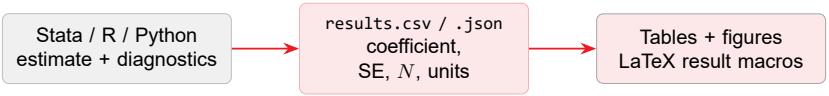
Public verification target: the theorem matches Mathlib's statement and depends only on Lean's three standard axioms, with no unfinished proof steps.

The new result is a computer-checkable version of established mathematics.

Anthropic (4 Sep. 2026); official repository, README and FinalCheck.lean. Status checked 24 Sep. 2026; build not rerun here.

Estimate once. Carry the number all the way through.

Statistical software → stored results → paper



One stored result feeds every appearance of the number in the paper.

Save locals (temporary values) and intermediate results to a file.

An agent can export estimates from Stata, R or Python, then fill tables, figures and LaTeX macros (named commands that insert results) from that same output.

Store the coefficient (estimated relationship), standard error (SE, a measure of uncertainty), sample size (N) and units. Rerun when the analysis changes.

Less manual copying. Fewer stale numbers and transcription errors.

Illustrative reproducible workflow. LaTeX can read generated result macros with `\input`; verify the exported quantities and compiled output.

Buy another round of criticism

Feedback throughout / Refine and Coarse

Refine.ink
A dedicated review and checking service: mathematical reasoning, consistency, references and claims.

Treat the report as a list of claims to investigate, not a grade to believe.

Coarse.ink
An automated referee-style review tool with public source code, model choice and direct API billing (paying per use).

Repeated, focused passes can be more useful than asking for one comprehensive blessing.

When should you ask for feedback?

Early, while the design is cheap to change?
Repeatedly throughout the project?
Only once something is polished?

Refine and coarse.ink product documentation, accessed 14 Sep. 2026. These are tools, not certifications of correctness.

Use the expensive model where it earns its keep

What feedback really costs

Dollars + tokens + rate limits + quotas

Tokens meter text in small chunks. Rate limits and quotas cap how much you can run.

Cheap, repeated passes
Extraction, formatting, broken references, simple consistency checks and initial screening.

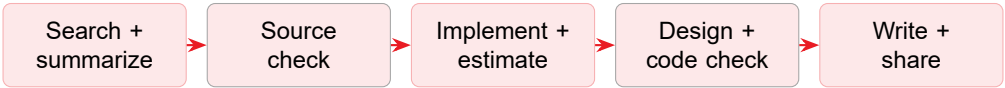
Use a stronger model when it matters
Difficult proofs, design criticism, conflicting evidence and consequential judgments.

In a Coarse/Refine-style review cycle, focus each pass and investigate the flagged issues. A model's additional capability should justify its effective cost.

Set a budget and a stopping rule for repeated reviews.

Coarse; Refine. Workflow guidance, not current price quotations. API charges and subscription limits are different constraints.

A faster workflow needs clear points to stop and check



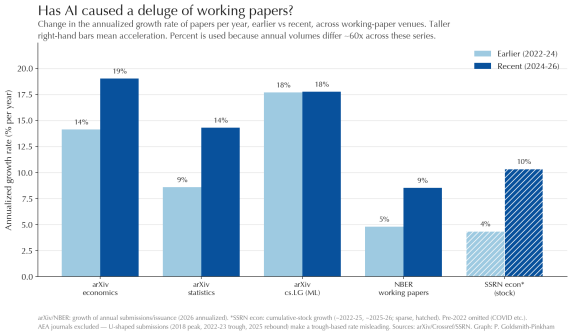
Before release: can someone else reconstruct the claim from the evidence?

Automate the work. Do not automate away the question “why should I believe this?”

Conceptual framework / practical workflow proposed for this talk.

A useful counterpoint: paper counts are not rising everywhere

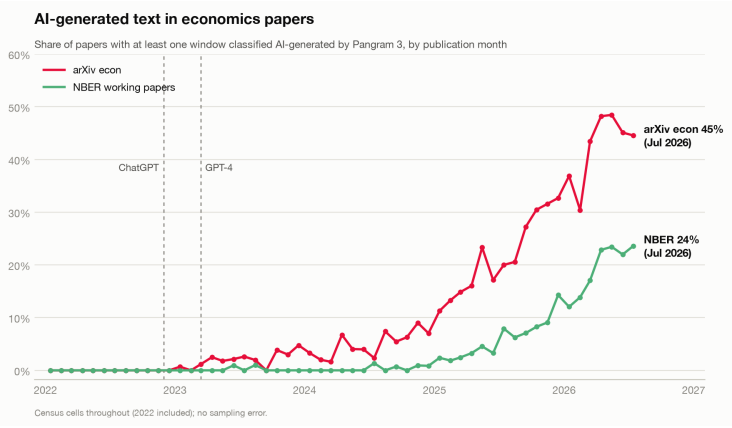
Paul Goldsmith-Pinkham / working papers: drafts shared before journal publication



Distinguish original papers from revisions and compare consistent periods.

Goldsmith-Pinkham (2026). Original growth-summary figure from the replication repository, retrieved 21 Sep. 2026.

The number of papers can stay stable while writing changes



45% arXiv econ

24% NBER

July 2026

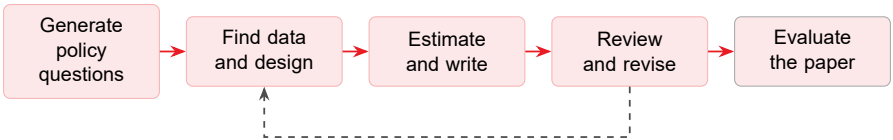
At least one sampled passage flagged by Pangram 3 (an AI-writing detector).

AI-like writing does not establish AI-generated research.

Goldsmith-Pinkham, How do we know if writing is AI? (10 Sep. 2026). Original figure; detector-dependent descriptive evidence.

Someone is already automating the whole pipeline

Project APE / Autonomous Policy Evaluation (AI-run policy research)



This is an attempt to automate a workflow, not merely to autocomplete a paragraph.

3,598 ideas

1,000 complete AI papers

18,000+ paper comparisons

Public project counts. Main production run frozen in April 2026.

Project APE, Social Catalyst Lab, University of Zurich. Website snapshot checked 14 Sep. 2026.

AI paper rankings are not review by experts

Project APE / output quality and its measurement

88.1%

reported probability that a human-written comparison paper beats an APE paper

Whose evaluation?
The project's LLM judge, in head-to-head paper comparisons.

What is being judged?
A manuscript, not necessarily whether its data, code or research design support the claims.

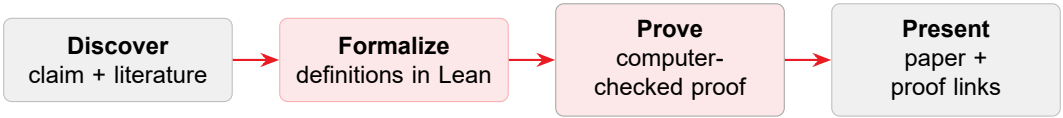
What follows?
Automating a paper and automating reliable knowledge are different targets.

How we measure quality becomes part of the research problem.

Project APE, Social Catalyst Lab, University of Zurich. Website snapshot checked 14 Sep. 2026.

CausalSmith automates a causal-theory workflow

Causal theory: mathematical rules for studying cause and effect



What runs automatically
Search existing research, propose claims, write them in Lean, attempt proofs and link the paper to that code.

What still needs judgment
Is the question useful? Do the definitions capture the causal idea? Are the assumptions credible in an application?

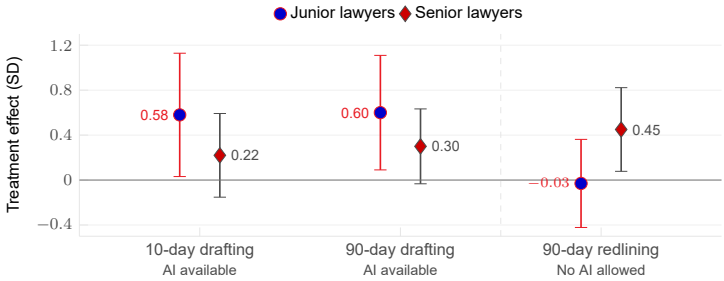
A larger research pipeline can be automated, with explicit checks between stages.

This develops causal-inference **theory**. It does not establish identification in an arbitrary dataset.

CausalSmith, How it works (checked 24 Sep. 2026). Simplified sourced workflow, redrawn for this talk.

AI can raise performance without building expertise

Autor et al. / three months of AI-assisted patent drafting



Assisted output and retained skill are different outcomes. Seniors improved on unassisted redlining (editing drafts); juniors showed no detectable average gain.

SD = standard deviation (a common scale for comparing scores).

Small subgroups; wide uncertainty. The difference between junior and senior drafting effects is not precisely established.

Autor et al. (2026), NBER WP 35720, Tables 3, 4 and 6, column 4. Expert ratings; bars show estimate +/- 1.96 SE.

Do not become a “meat proxy"

meat proxy *noun*

A human who passes an AI’s output to another person without adding judgment, verification, interpretation or value.

Working definition, paraphrasing Niklas Gruhn.

Usage: “I asked what you think, not what the chatbot thinks.”

If my name is on the answer, I should be able to explain and defend it.

Niklas Gruhn, Don't be a meat proxy (3 Aug. 2026). Definition and illustrative usage adapted for this talk, not verbatim quotations. Original repost retained in the asset folder.

Don't throw “slop grenades"

slop grenade *noun*

A blast of unchecked AI output that saves the sender time by making everyone else do the sorting and checking.

Working definition / paraphrase.

Research example: a 20-page AI referee report, forwarded unread.

Read it. Check it. Find the point. Then decide what is worth sending.

Popularized by Tobi Lütke. His credit to Harry Brundage led back to Stefan Neubig’s noslopgrenade.com.

X, 15 Sep. 2026: Parrish; Lütke; Brundage's correction; noslopgrenade.com. Cited attribution chain, not a claim of first-ever use.

Do not build a private reality with a chatbot

Caution / sycophancy (telling you what you want to hear), overreliance and “AI psychosis”

A research warning sign
Every new objection somehow confirms that the project is brilliant, revolutionary and misunderstood.

What counts as evidence
A model’s confidence, praise or apparent intimacy is not independent evidence.

Keep outside checks
Colleagues, source documents, checks others can repeat and time away from the conversation.

Keep people in the picture
Serious distress or loss of perspective calls for human support, not a longer argument with the model.

Use the system as a tool, not as the sole judge of your ideas — or your reality.

OpenAI (2025), sensitive conversations. Risk discussion; not an estimate of clinical incidence or a causal claim.

My AI risk-mitigation strategy

The prompt behind this image

“Based on our conversation history, create a picture of how you feel I treat you.”



An AI Pascal’s Wager

I’m not saying the machines will take over.

I’m just saying I’d prefer the logs to show I was courteous.

Source: user-uploaded AI-generated image. Full image prompt shown above.

Where does human value move?

The Industrial Revolution in the Idea Factory

Choose

Questions worth answering.
Claims worth making.

Check

Designs, data, calculations.
Meaning, not just correct-looking symbols and code.

Own

The explanation.
The uncertainty.
The final product.

If research production becomes dramatically cheaper, what becomes scarce and valuable?

Discussion

Conceptual framework / practical workflow proposed for this talk.